Analytical Workflows

IB 516

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Overview

Overview

Course Principles

Automation

Doing science involves repetition at multiple levels.
Automation is more efficient and reduces mistakes.

Reproducibility

Science must be reproducible, by others and you.
Your work should be *transparent*, *organized*, and *self-contained*.

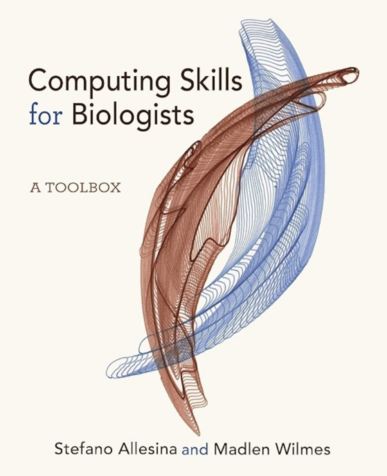
Correctness

Science is done by humans; mistakes are inevitable.
We must work in ways that protect against bugs.

[1] See Allesina & Wilmes 2019

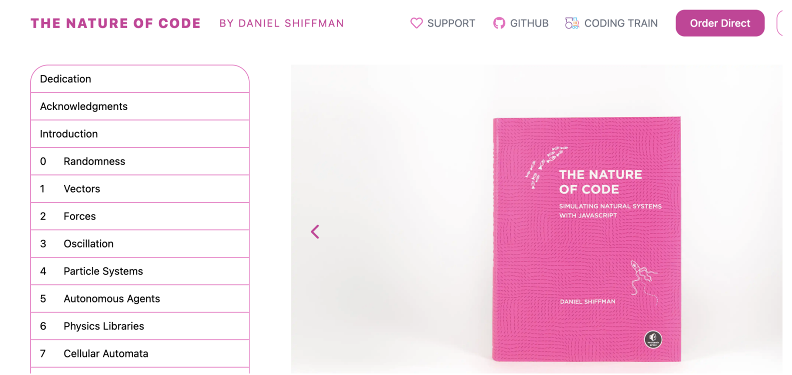
Not required reading

But good! Currently missing from the Valley library...



Further optional reading

Helpful for seeing how pipelines are built step by step. Also really fun. Lots online: `natureofcode.com`



Course Principles

Simplicity

Analyses should be coded as simply as possible.
The clearest solution is often the best one.

Openness

Science is a worldwide endeavor.
Access to it should be free and equitable.

Science as Software Development

Science is a *process* of versions and updates.
Software developers have tools to help.

[1] See Allesina & Wilmes 2019

Overview

Website & Schedule

`github.com/analyticalworkflows/TeachingMaterials`

Accountabilibuddies

- Weekly check-in
  - *goals/deadlines, accomplishments, sticking-points*
  - *final 10 min. of every 2nd class*
- Randomized pairing
  - *re-randomized each week*

Overview

What you'll need

Accounts

- GitHub[1]

Software

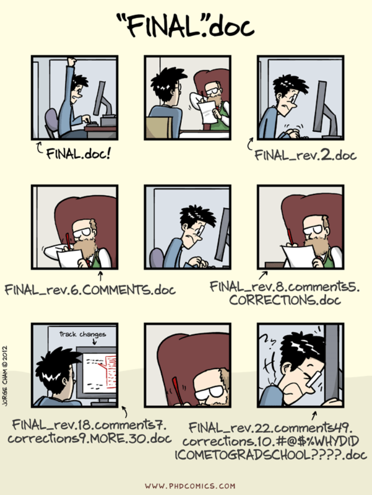
- An integrated development environment (IDE), such as Visual Studio Code (VS Code), RStudio, etc.

Git interface

- GitHub Desktop, VS Code, RStudio, Sourcetree, etc.

LaTeX editor

- Stand-alone: TeXstudio, VS Code, etc.
- Online: Overleaf, Papeeria, TeXpage, etc.



[1] Please email me your username.

Overview

Project Proposals

Primary Goal

To develop more effective research skills.

Secondary Goal

To make significant progress on your thesis.

Our belief

We can achieve both in this course!

Let's get started!
.../TeachingMaterials/classes/ProjectProposal

Overview

We're all learning

- Use this course as a workshop for your thesis
- Lean into the work—you and your peers will benefit
- Be kind to yourself
- Support each other